

Throat-Geometry Overlap for Alpha Preformation: Warped Measure and Woods–Saxon Cluster Trial

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Abstract

The flat $[0, R]$ standing-wave overlap proxy [2] uses $\ell = 0$ flag mode vs Gaussian alpha trial with measure dr . This sequel replaces that proxy geometry with (i) the AdS-like throat radial weight $w(r) = R/r$ implied by the companion nuclear metric $ds^2 = (R^2/r^2)dr^2 + \dots$ [1], and (ii) a Woods–Saxon (WS) alpha-cluster radial trial as primary, keeping Gaussian/HO as comparison. Overlap support remains the confined interior $[0, R]$ where the Dirichlet flag mode is defined; the barrier $[R, b]$ stays the Gamow domain. On the same 14-isotope set, primary throat+WS yields $\langle \log_{10} P_{\text{model}} \rangle = -0.4607$, Pearson $r = +0.8875$ vs extracted $\log_{10} P_\alpha$, and $S_\star \approx 3.18 \times 10^{-2}$ reducing RMS residual from 1.7099 to 0.8247. Relative to the flat-Gaussian proxy, correlation changes by $\Delta r \approx -0.0007$; absolute-scale S shifts modestly. Still a geometric/structural proxy—not microscopic uniqueness.

1 Motivation and what moved

Reference [2] documented a flat-measure overlap proxy and flagged full throat geometry and a more realistic alpha trial as the next structural step. This note implements that step on the unchanged 14-isotope table of [1].

Remark (RC1 scope). P_{model} remains a geometric/structural preformation *proxy*. Warped measure and WS shape are mechanism choices documented below; they are not a claim of unique many-body nuclear structure. Absolute scale may still need one global factor S .

1.1 What moved (proxy $\rightarrow$ throat/WS)

- **Measure:** flat $dr \rightarrow$ throat weight $w(r) = R/r$ from $\sqrt{g_{rr}}$ on the AdS-like confining metric of [1].
- **Alpha trial:** Gaussian/HO $\rightarrow$ Woods–Saxon primary $u_\alpha(r) = r(1 + e^{(r-R_\alpha)/a_{ws}})^{-1}$, with optional WS $\times$ Coulomb-tail match for comparison.
- **Artifacts:** new code `palpha_overlap_throat.py`, JSON `palpha_overlap_throat_results.json`; baseline proxy files retained.

1.2 What remains open

- Isotope-specific spectroscopic amplitudes beyond one global S .
- Self-consistent bound-state solution of WS+Coulomb (R-matrix / cluster eigenmode) rather than a phenomenological trial.
- Overlap constructions that place non-vanishing flag amplitude in the barrier $[R, b]$ (the present Dirichlet u_{in} vanishes at R).
- Absolute uniqueness of preformation from many-body structure.

Remark (Next). Parametric isotope-resolved $S(A, Z)$ (with leave-one-out) and a self-consistent WS+Coulomb interior eigenmode are implemented as comparison channels in the refined sequel `flag_condensate_palpa_refined.` / `palpa_overlap_refined.py`. On the stated set, parametric $S(A, Z)$ reduces LOO residual RMS most among non-tautological protocols; see that note for metrics.

2 Throat weight and overlap definition

Companion metric [1]:

$$ds^2 = \frac{R^2}{r^2} dr^2 + \frac{r^2}{R^2} \eta_{\mu\nu} dx^\mu dx^\nu. \quad (1)$$

We take the radial measure factor $\sqrt{g_{rr}} = R/r$ and define

$$P_{\text{model}} = \frac{\left| \int_0^R u_{\text{in}}(r) u_\alpha(r) w(r) dr \right|^2}{\left(\int_0^R u_{\text{in}}^2 w dr \right) \left(\int_0^R u_\alpha^2 w dr \right)}, \quad w(r) = \frac{R}{r}. \quad (2)$$

For reduced modes $u \sim O(r)$ near the origin the integrands remain integrable. Flat $w \equiv 1$ is recomputed for side-by-side comparison (including reproduction of [2]).

Remark (Domain choice). Overlap support is $[0, R]$. The Coulomb barrier interval $[R, b]$ determines Gamow W in [1]; it is not used here as overlap support for the Dirichlet interior mode.

3 Radial modes

3.1 Confined flag mode ($\ell = 0$)

Unchanged:

$$u_{\text{in}}(r) = r j_0(kr), \quad k = \pi/R. \quad (3)$$

3.2 Alpha trials

Primary Woods–Saxon cluster packet (inputs $r_0 = 1.2$ fm, $a_{\text{WS}} = 0.55$ fm):

$$u_\alpha^{\text{WS}}(r) = \frac{r}{1 + e^{(r-R_\alpha)/a_{\text{WS}}}}, \quad R_\alpha = r_0 4^{1/3} \approx 1.905 \text{ fm}. \quad (4)$$

Comparison: Gaussian/HO as in [2]; optional WS with exponential $\times R_m/r$ tail matched in value at $R_m = R_\alpha + 2a_{\text{WS}}$ ($\kappa = 1/a_{\text{WS}}$, phenomenological).

4 Numerical protocol

Simpson quadrature, $N = 4096$ on $[0, R]$. Implementation: `rh_papers_may21/acs-framework/code/palpa_overlap`. Results: `docs/palpa_overlap_throat_results.json`. Logs: `docs/palpa_overlap_throat_logs/`.

5 Results on the stated set

Table 1: **Throat+WS overlap vs extracted $\log_{10} P_\alpha$.** Primary model; S is one global factor.

Isotope	R (fm)	$\log_{10} P_{\text{model}}$	$\log_{10} P_\alpha^{\text{ext}}$	resid. (raw)	$\log_{10}(SP_{\text{model}})$	resid. (S)
^{212}Po	8.99	-0.4441	-1.53	+1.0859	-1.9419	-0.4119
^{214}Po	9.02	-0.4465	-1.19	+0.7435	-1.9443	-0.7543
^{216}Po	9.05	-0.4489	-1.04	+0.5911	-1.9467	-0.9067
^{218}Po	9.07	-0.4505	-0.82	+0.3695	-1.9483	-1.1283
^{220}Rn	9.10	-0.4529	-1.12	+0.6671	-1.9507	-0.8307
^{224}Ra	9.16	-0.4576	-1.54	+1.0824	-1.9555	-0.4155
^{226}Ra	9.18	-0.4592	-1.73	+1.2708	-1.9571	-0.2271
^{228}Th	9.21	-0.4616	-1.87	+1.4084	-1.9595	-0.0895
^{232}Th	9.26	-0.4656	-2.17	+1.7044	-1.9634	+0.2066
^{234}U	9.29	-0.4679	-2.64	+2.1721	-1.9658	+0.6742
^{238}U	9.34	-0.4718	-1.97	+1.4982	-1.9697	+0.0003
^{238}Pu	9.34	-0.4718	-3.28	+2.8082	-1.9697	+1.3103
^{240}Pu	9.36	-0.4734	-2.91	+2.4366	-1.9713	+0.9387
^{244}Cm	9.42	-0.4781	-3.61	+3.1319	-1.9760	+1.6340

5.1 Primary summary (throat + Woods–Saxon)

$$\begin{aligned}
\langle \log_{10} P_{\text{model}} \rangle &= -0.4607 \quad (\sigma = 0.0110), \\
\langle \log_{10} P_\alpha^{\text{ext}} \rangle &= -1.9586 \quad (\sigma = 0.8656), \\
r(\log_{10} P_{\text{model}}, \log_{10} P_\alpha^{\text{ext}}) &= +0.8875, \\
\text{RMS}_{S=1} &= 1.7099, \quad S_\star = 3.178 \times 10^{-2}, \quad \text{RMS}_{S_\star} = 0.8247.
\end{aligned}$$

5.2 Model comparison on the stated set

Table 2: Flat vs throat measure; Gaussian vs WS alpha trial.

Model	$\langle \log_{10} P \rangle$	r	$\text{RMS}_{S=1}$	$\text{RMS}_{S_\star}$
throat + WS (primary)	-0.4607	+0.8875	1.7099	0.8247
throat + WS×Coulomb tail	-0.4825	+0.8875	1.6908	0.8245
throat + Gaussian	-0.2791	+0.8881	1.8715	0.8260
flat + WS	-0.6786	+0.8876	1.5201	0.8200
flat + Gaussian (baseline proxy)	-0.3905	+0.8882	1.7705	0.8220

Relative to flat+Gaussian [2]: $\Delta \langle \log_{10} P \rangle = -0.0702$, $\Delta r = -0.0007$, $\Delta \text{RMS}_{S_\star} = +0.0027$, S moves from 2.70×10^{-2} to 3.18×10^{-2} .

Remark (Interpretation). On this set, warped measure and WS shape shift absolute scale and slightly compress $\sigma(\log_{10} P_{\text{model}})$, but do not materially change correlation with extracted $\log_{10} P_\alpha$. Isotope-specific residual structure remains; one global S absorbs mean offset only.

6 Conclusion

We have advanced the flat $[0, R]$ proxy of [2] to an AdS-like throat weight $w = R/r$ with Woods–Saxon alpha trial on the same 14-isotope set. Primary metrics: $\langle \log_{10} P \rangle = -0.4607$, $r = +0.8875$, $S_\star \approx 0.032$. Correlation is essentially unchanged versus the flat-Gaussian proxy. Isotope-resolved / parametric S and self-consistent WS+Coulomb eigenmodes are reported in the refined sequel [3]. Absolute uniqueness is not claimed.

References

- [1] ACS Theoretical Physics Working Group, *Nuclear Decay as a Spectral Bifurcation of the Flag Condensate*, ACS Framework note (2026), `flag_condensate_nuclear_decay.tex` / `papers/notes/Flag_Condensate_Nuclear_Decay.tex`.

- [2] ACS Theoretical Physics Working Group, *Standing-Wave Overlap Proxy for Alpha Preformation*, ACS Framework note (2026), `flag_condensate_palpha_overlap.tex` / `papers/notes/Flag_Condensate_Palpha_Overlap.tex`.
- [3] ACS Theoretical Physics Working Group, *Refined P_α Overlap: Parametric S and $WS+Coulomb$ Eigenmode*, ACS Framework note (2026), `flag_condensate_palpha_refined.tex` / `papers/notes/Flag_Condensate_Palpha_Refined.tex`.
- [4] G. Gamow, *Zur Quantentheorie des Atomkernes*, Z. Phys. **51**, 204 (1928).